Sahaj Bhadja

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Summary

Intelligent Transportation Systems Engineer with 2+ years building **Python/C++ systems**, **autonomous-driving simulations**, system integration, and **validation workflows** using spatial modeling, algorithms, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **system validation**, reconciliation, error handling, cloud storage integration, CI/CD releases, and stable production monitoring workflows reliably at enterprise scale.
- Reduced dashboard latency from 1.2s to 350ms through **data modeling and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured evaluation workflows**, model switching, fallback paths, persistent results, and reviewer controls supporting repeatable analytical processing for technical users.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **system integration and debugging**, indexing, caching, pagination, and filtered payloads across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship | *India*
- Reduced centralized-control and quantum-risk exposure by 25% through a **novel consensus algorithm**, comparing benchmark behavior, performance trade-offs, and failure characteristics against established distributed mechanisms systematically and reproducibly.
- Published peer-reviewed **systems research** and presented at IISc Bengaluru through IEEE venues, documenting architecture, simulation methodology, threat models, experimental trade-offs, and reproducible findings for technical and non-technical audiences.

Skills

Languages: Python, C++, Java, SQL, JavaScript, Bash
Transport Modeling & AI: Trajectory Planning, HD Maps, Spatial-Temporal Modeling, Autonomous Driving, Simulation, Decision Transformers, Offline Reinforcement Learning, PyTorch, Transformers, Attention, Model Evaluation
Systems & Integration: System Design, System Integration, System Validation, Interoperability Testing, REST APIs, Microservices, Distributed Systems, Concurrency, Multithreading, Fault Tolerance
Data & Analysis: PostgreSQL, Data Modeling, Query Optimization, Data Pipelines, Feature Engineering, Data Visualization
Tools & Delivery: Linux, Windows, Git, GitHub, Jenkins, Docker, Kubernetes, AWS, Azure, CI/CD
Engineering Practices: Data Structures, Algorithms, Object-Oriented Programming, Unit Testing, Integration Testing, Automated Testing, PyTest, Debugging, Monitoring, Performance Optimization, Technical Documentation, Research Methods, Agile

Projects

Connected Corridor Operations Simulator - Design Concept
- Designed a **connected-corridor simulation** concept using Python, map-based trajectories, traffic-state scenarios, configurable roadside assets, incident events, operational KPIs, comparative deployment analysis, and reproducible evaluation reports for planning decisions.

ITS Requirements and Validation Workbench - Design Concept
- Designed an **ITS validation workbench** concept linking Concept of Operations scenarios, stakeholder requirements, interface definitions, verification procedures, expected results, issue tracking, design reviews, and traceable user-validation evidence.

CPU-Scheduling Simulator | Live
- Built a **C++ systems simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across operational scenarios.